

The Plane-Wave Response of a Sphere

The angular spectrum of the elastic Mie field, and the reflection and transmission matrices on planes bounding the scatterer

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Abstract

A homogeneous elastic sphere in a homogeneous whole space scatters an incident plane wave into a field whose exact solution is a series of spherical waves. This note gives that field's *plane-wave* content: the amplitude, at each lateral wavenumber, of every up- and down-going P , SV and SH wave crossing a plane placed just outside the sphere. Those amplitudes are the reflection and transmission matrices $\mathbf{R}(\mathbf{k}_\perp)$ and $\mathbf{T}(\mathbf{k}_\perp)$.

The construction is closed-form. Each outgoing multipole has a plane-wave representation whose angular weight is its own spherical harmonic evaluated at the plane-wave direction, so the spectrum of the scattered field is read off its Mie coefficients with no spatial transform, no lateral window and no truncation. Everything needed to implement it is given: the potentials and their coefficients, the displacement and traction, the plane-wave representation and its branch, the three channel spectra with their polarisations, the quadrature path that removes the branch-point singularity, and the assembly into $\mathbf{R}$ and $\mathbf{T}$.

Conventions are fixed in §1.1 and used throughout: time dependence $e^{-i\omega t}$, outgoing radial functions $h_n^{(1)}$, and $\text{Im } k_z \geq 0$.

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1 Statement of the problem

A sphere of radius a , with Lamé parameters λ_1, μ_1 and density ρ_1 , sits in an unbounded homogeneous medium λ, μ, ρ with wave speeds

$$\alpha = \sqrt{(\lambda + 2\mu)/\rho}, \quad \beta = \sqrt{\mu/\rho}, \quad k_P = \omega/\alpha, \quad k_S = \omega/\beta. \quad (1)$$

A plane wave is incident along $+z$. Take two planes $z = -z_p$ and $z = +z_p$ with $z_p > a$, so that both lie wholly outside the sphere.

We want, at each lateral wavenumber $\mathbf{k}_\perp = (k_x, k_y)$, the amplitudes of the plane waves crossing those planes. Above the sphere every scattered wave travels in $-z$; below, in $+z$. Collecting the three polarisations gives a 3×3 reflection matrix on the upper plane and a 3×3 transmission matrix on the lower one, each a function of $\mathbf{k}_\perp$.

1.1 Conventions

- Time dependence $e^{-i\omega t}$; outgoing spherical waves carry $h_n^{(1)}(kr)$.
- z increases downward and is the first Cartesian index, so a vector is ordered (z, x, y) . Spherical angles are measured from $+z$: $\cos \theta = z/r$, and φ is the azimuth in the (x, y) plane.
- The vertical wavenumber of a plane wave is taken on the branch

$$k_z(\mathbf{k}_\perp) = \sqrt{k^2 - |\mathbf{k}_\perp|^2}, \quad \text{Im } k_z \geq 0, \quad (2)$$

real inside the light circle $|\mathbf{k}_\perp| < k$ and positive-imaginary outside it, so that $e^{ik_z|z|}$ is outgoing in both half-spaces: propagating within the circle and decaying beyond it.

- Seismic units are used throughout — km/s, g/cm³, GPa, km. In SI the matrices of §6 carry displacement rows of order unity against traction rows of order $\mu k \sim 10^6$, and appear ill conditioned for a reason that is purely one of units.

2 The exact field in spherical waves

2.1 Potentials

Write the displacement through the Helmholtz decomposition

$$\mathbf{u} = \nabla\phi + \nabla \times \nabla \times (\mathbf{r}\psi) + \nabla \times (\mathbf{r}\chi), \quad (3)$$

where $\mathbf{r}$ is the position vector and each potential satisfies a scalar Helmholtz equation: ϕ with k_p , and ψ, χ with k_s . The three terms are the L -, N - and M -type families, radiating P , SV and SH respectively. The two shear terms are divergence free, so

$$\nabla \cdot \mathbf{u} = \nabla^2 \phi = -k_p^2 \phi. \quad (4)$$

Outside the sphere the scattered potentials are outgoing:

$$\phi = \sum_n a_n h_n^{(1)}(k_p r) P_n^m(\cos \theta) e^{im\varphi}, \quad \psi = \sum_n b_n h_n^{(1)}(k_s r) P_n^m(\cos \theta) e^{im\varphi}, \quad (5)$$

and likewise χ with coefficients c_n . Inside, the same forms with j_n and the interior wavenumbers.

An axial P wave is azimuthally symmetric, so it excites only $m = 0$, and it excites no toroidal field: $c_n = 0$. Incident SV and SH excite $m = 1$. The channel formulas of §4 cover both, with the angular functions (21) giving the $m = 0$ and $m = 1$ cases explicitly.

2.2 The incident field

The incident wave enters twice, in different forms, and both are needed.

As a series, to drive the boundary conditions. A unit-amplitude P wave travelling along $+z$ is $\mathbf{u}^{\text{inc}} = \hat{\mathbf{z}} e^{ik_p z}$, which comes from the potential $\phi^{\text{inc}} = e^{ik_p z}/(ik_p)$ since $\mathbf{u} = \nabla\phi$. The Rayleigh expansion

$$e^{ikz} = \sum_{n \geq 0} (2n+1) i^n j_n(kr) P_n(\cos \theta) \quad (6)$$

then gives the incident partial-wave coefficients

$$a_n^{\text{inc}} = \frac{(2n+1) i^n}{ik_p}, \quad b_n^{\text{inc}} = c_n^{\text{inc}} = 0, \quad (7)$$

so axial P incidence is purely $m = 0$ and purely L -type.

A unit shear wave along $+z$ is transverse, $\mathbf{u}^{\text{inc}} = \hat{\mathbf{x}}e^{ik_{sz}}$ for SV and $\hat{\mathbf{y}}e^{ik_{sz}}$ for SH . Its expansion carries P_n^1 and $e^{\pm i\varphi}$, so shear incidence is $m = 1$, with the N -type and M -type potentials respectively. The conversion from the $m = 0$ plane-wave coefficients to the $m = 1$ ones is

$$\rho(n) = -\frac{1}{n(n+1)}, \quad (8)$$

which carries the normalisation of P_n^1 relative to P_n .

As a plane wave, in the angular spectrum. The same incident field is, trivially, a single plane wave, so its spectrum is a delta:

$$\hat{\mathbf{u}}^{\text{inc}}(\mathbf{k}_{\perp}) = (2\pi)^2 \mathbf{e}_i \delta^2(\mathbf{k}_{\perp} - \mathbf{k}_{\perp}^0), \quad (9)$$

with $\mathbf{k}_{\perp}^0 = \mathbf{0}$ for axial incidence. It travels in $+z$, so it is present on the lower plane and absent on the upper one. That is the whole of the difference between $\mathbf{R}$ and $\mathbf{T}$ in §6: reflection is purely scattered, transmission is the incident delta plus the scattered part.

2.3 Coefficients

The a_n, b_n follow from continuity of $u_r, u_{\theta}, \sigma_{rr}$ and $\sigma_{r\theta}$ at $r = a$ — a 4×4 system per order, with columns (scattered P , scattered S , interior P , interior S) and right-hand side built from (7), or its shear counterpart through (8). The c_n follow from the corresponding 2×2 system in u_{φ} and $\sigma_{r\varphi}$. These are the classical Mie relations and are not restated here.

2.4 Displacement

For $m = 0$ the L - and N -type fields have no azimuthal component, and with $z_n \equiv h_n^{(1)}$,

$$\text{L-type: } u_r = a_n k_p z'_n(k_p r) P_n, \quad u_{\theta} = a_n \frac{z_n(k_p r)}{r} \frac{dP_n}{d\theta}, \quad (10)$$

$$\text{N-type: } u_r = b_n n(n+1) \frac{z_n(k_s r)}{r} P_n, \quad u_{\theta} = b_n \left[\frac{z_n(k_s r)}{r} + k_s z'_n(k_s r) \right] \frac{dP_n}{d\theta}. \quad (11)$$

The M -type is purely azimuthal,

$$u_r = u_{\theta} = 0, \quad u_{\varphi} = -c_n z_n(k_s r) \frac{dP_n}{d\theta}. \quad (12)$$

Throughout, $dP_n/d\theta = P_n^1(\cos \theta)$ in the Condon–Shortley convention, which follows from $P_n^1(u) = -\sqrt{1-u^2} P'_n(u)$ and $dP_n/d\theta = -\sin \theta P'_n(u)$, the two coinciding because $\sin \theta \geq 0$ on $[0, \pi]$. The combination $P_n^1(\cos \theta)/\sin \theta = -P'_n(\cos \theta)$ is finite on the axis, with limit $-n(n+1)/2$.

2.5 Traction on a plane of constant z

The state used in §6 is $(\mathbf{u}, \boldsymbol{\tau}_3)$ with $\boldsymbol{\tau}_3 = \boldsymbol{\sigma} \cdot \hat{\mathbf{z}}$. From $\boldsymbol{\sigma} = \lambda(\nabla \cdot \mathbf{u})\mathbf{I} + \mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$,

$$(\boldsymbol{\sigma} \cdot \hat{\mathbf{z}})_i = \lambda(\nabla \cdot \mathbf{u}) \delta_{iz} + \mu(\partial_z u_i + \partial_i u_z), \quad (13)$$

so only the z-row and z-column of the displacement gradient are required, and $\nabla \cdot \mathbf{u}$ is known in closed form from (4).

For $m = 0$ the field is axisymmetric with no azimuthal component, so $\sigma_{r\varphi} = \sigma_{\theta\varphi} = 0$ and the strain reduces to

$$\begin{aligned} \varepsilon_{rr} &= \partial_r u_r, & \varepsilon_{\theta\theta} &= \frac{\partial_\theta u_\theta + u_r}{r}, \\ \varepsilon_{\varphi\varphi} &= \frac{u_r + \cot \theta u_\theta}{r}, & \varepsilon_{r\theta} &= \frac{1}{2} \left[\frac{\partial_\theta u_r}{r} + \partial_r u_\theta - \frac{u_\theta}{r} \right]. \end{aligned} \quad (14)$$

Each term is finite on the axis: u_θ carries $dP_n/d\theta \sim \sin \theta$, so $\cot \theta u_\theta$ is regular and is evaluated through $P_n^1/\sin \theta$ rather than by dividing. Legendre's equation removes the second angular derivative,

$$\frac{d^2 P_n}{d\theta^2} = -\cot \theta \frac{dP_n}{d\theta} - n(n+1)P_n, \quad (15)$$

so no function beyond P_n and P_n^1 is needed. With $\sigma = \lambda \operatorname{tr}(\varepsilon) \mathbf{I} + 2\mu \varepsilon$ in the orthonormal spherical basis, and $\hat{\mathbf{r}} \cdot \hat{\mathbf{z}} = \cos \theta$, $\hat{\boldsymbol{\theta}} \cdot \hat{\mathbf{z}} = -\sin \theta$, $\hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}} = 0$,

$$\boldsymbol{\tau}_3 = (\sigma_{rr} \cos \theta - \sigma_{r\theta} \sin \theta) \hat{\mathbf{r}} + (\sigma_{r\theta} \cos \theta - \sigma_{\theta\theta} \sin \theta) \hat{\boldsymbol{\theta}}. \quad (16)$$

The radial derivatives in (14) follow from (10)–(12) with $z_n'' = (n(n+1)/x^2 - 1)z_n - (2/x)z_n'$, the spherical Bessel equation.

3 The plane-wave representation

3.1 The identity

Every outgoing multipole is a superposition of plane waves. For $z > 0$,

$$h_n^{(1)}(Kr) Y_n^m(\hat{\mathbf{r}}) = \frac{1}{2\pi K i^n} \iint dk_x dk_y e^{i\mathbf{k} \cdot \mathbf{r}} \frac{Y_n^m(\hat{\mathbf{k}})}{k_z}, \quad \mathbf{k} = (k_z, k_x, k_y), \quad (17)$$

with k_z on the branch (2). For $z < 0$ replace k_z by $-k_z$ in $\mathbf{k}$ and in $\hat{\mathbf{k}}$, so that the exponential is $e^{i(k_x x + k_y y + k_z |z|)}$ in either half-space. At $n = 0$, (17) is the classical Weyl identity for h_0 .

The angular weight is the multipole's own harmonic *evaluated at the plane-wave direction*. This is what makes the spectrum closed-form: no transform of the field is involved.

3.2 The harmonic must be the solid form

Outside the light circle k_z is imaginary, so $\hat{\mathbf{k}}$ is a complex direction and its polar angle is complex. Y_n^m must therefore be evaluated in a form that is analytic in its arguments — the solid harmonic, a homogeneous degree-zero ratio of polynomials in (k_x, k_y, k_z) . A representation through real polar angles is defined only inside the light circle and silently discards the evanescent band, which is most of the domain of integration.

In practice this means evaluating P_n , P_n' and P_n'' at the complex argument k_z/K , which (21) requires; all three are polynomials and take a complex argument without qualification.

4 The three channel spectra

4.1 Polarisations

Write $q = |\mathbf{k}_\perp|$, let ψ_k be the azimuth of $\mathbf{k}_\perp$, and let $k_z^\pm$ denote $+k_z$ below the sphere and $-k_z$ above it. At each $\mathbf{k}_\perp$ the scattered field is a sum of exactly three plane waves: one P at $k_{z,P}$ and two shear at $k_{z,S}$. Their polarisations are

$$\mathbf{e}_P = \frac{(k_z^\pm, k_x, k_y)}{k_P}, \quad \mathbf{e}_{SV} = \frac{(-q, k_z^\pm \cos \psi_k, k_z^\pm \sin \psi_k)}{k_S}, \quad \mathbf{e}_{SH} = (0, -\sin \psi_k, \cos \psi_k), \quad (18)$$

each of unit length, with $\mathbf{e}_{SV}$ and $\mathbf{e}_{SH}$ orthogonal to $\mathbf{k}$. They depend on the direction only, not on n or m .

4.2 The angular function and its surface gradient

Let A_{nm} be the angular function of the potential — $P_n(\cos \theta)$ at $m = 0$, $P_n^1(\cos \theta) \cos \varphi$ at $m = 1$ — evaluated at the *plane-wave direction*, so that $\cos \theta \rightarrow k_z^\pm/K$, $\sin \theta \rightarrow q/K$ and $\varphi \rightarrow \psi_k$. The two transverse vector harmonics built from it are

$$\nabla_s A = \frac{\partial A}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{\sin \theta} \frac{\partial A}{\partial \varphi} \hat{\boldsymbol{\phi}}, \quad \hat{\mathbf{r}} \times \nabla_s A = -\frac{1}{\sin \theta} \frac{\partial A}{\partial \varphi} \hat{\boldsymbol{\theta}} + \frac{\partial A}{\partial \theta} \hat{\boldsymbol{\phi}}, \quad (19)$$

with $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$ at the plane-wave direction identified with $\mathbf{e}_{SV}$ and $\mathbf{e}_{SH}$ respectively.

4.3 The spectra

With the normalisations of (17) cancelling, and writing $D = 2\pi K i^n k_z$ with $K = k_P$ for the L -type and k_S for the other two,

$$\begin{array}{ll} L\text{-type } (a_n) : & \frac{ik_P}{D} A_{nm} \mathbf{e}_P, \\ N\text{-type } (b_n) : & \frac{ik_S}{D} \left[\frac{\partial A_{nm}}{\partial \theta} \mathbf{e}_{SV} + \frac{1}{\sin \theta} \frac{\partial A_{nm}}{\partial \varphi} \mathbf{e}_{SH} \right], \\ M\text{-type } (c_n) : & \frac{-1}{D} \left[-\frac{1}{\sin \theta} \frac{\partial A_{nm}}{\partial \varphi} \mathbf{e}_{SV} + \frac{\partial A_{nm}}{\partial \theta} \mathbf{e}_{SH} \right], \end{array} \quad (20)$$

each multiplied by its Mie coefficient and summed over n . The three constants ik_P , ik_S and -1 are independent of n and of m .

Explicitly, with $u = k_z^\pm/K$ and $s = q/K$,

$$\begin{array}{lll} m = 0 : & A = P_n(u), & \frac{\partial A}{\partial \theta} = -s P_n'(u), \quad \frac{1}{\sin \theta} \frac{\partial A}{\partial \varphi} = 0, \\ m = 1 : & A = -s P_n'(u) \cos \psi_k, & \frac{\partial A}{\partial \theta} = [-u P_n'(u) + s^2 P_n''(u)] \cos \psi_k, \quad \frac{1}{\sin \theta} \frac{\partial A}{\partial \varphi} = P_n'(u) \sin \psi_k. \end{array} \quad (21)$$

4.4 Three remarks

The L -type is a multiplier; the others are not. $\mathbf{u} = \nabla\phi$ gives the P amplitude immediately. $\nabla \times \nabla \times (\mathbf{r}\psi)$ carries an explicit position vector, and acting on a single plane-wave component,

$$\nabla \times \nabla \times (\mathbf{r} e^{i\mathbf{k}\cdot\mathbf{r}}) = e^{i\mathbf{k}\cdot\mathbf{r}} [k_S^2 \mathbf{r} - \mathbf{k}(\mathbf{k} \cdot \mathbf{r})] - 2i\mathbf{k} e^{i\mathbf{k}\cdot\mathbf{r}}, \quad (22)$$

so the $\mathbf{r}$ -dependence cancels only under the $\mathbf{k}$ integration and not term by term. The results in (20) are nonetheless exact.

Each shear family feeds *both* transverse polarisations. That is the content of (19), and it is why (20) cannot be written as one amplitude per family. At $m = 0$ the azimuthal derivative vanishes, each family collapses onto a single polarisation, and the structure is invisible — which is the case most often written down.

The M -type constant carries no factor of k . $\nabla \times \nabla \times (\mathbf{r} \cdot)$ has one derivative more than $\nabla \times (\mathbf{r} \cdot)$, so potentials normalised alike give amplitudes differing by a power of k_S : the constants are ik_S and -1 respectively.

5 Evaluating the representation

The spectra (20) are what $\mathbf{R}$ and $\mathbf{T}$ are built from, and need no integration. Reconstructing the field from them — to verify the construction, or to obtain the field at a point — requires the integral in (17), and two reductions make it straightforward.

5.1 The azimuthal integral, in closed form

For $m = 0$ the only azimuthal dependence of the integrand is the plane-wave phase and the polarisation. With $\rho = \sqrt{x^2 + y^2}$ and φ its azimuth,

$$\int_0^{2\pi} e^{iq\rho \cos(\psi_k - \varphi)} d\psi_k = 2\pi J_0(q\rho), \quad \int_0^{2\pi} \cos \psi_k e^{iq\rho \cos(\psi_k - \varphi)} d\psi_k = 2\pi i \cos \varphi J_1(q\rho), \quad (23)$$

and similarly with $\sin \psi_k \rightarrow \sin \varphi$. The double integral collapses to a one-dimensional integral in q . The vertical component of the P and SV fields pairs with J_0 and the horizontal with J_1 ; the SH field, having no vertical component, carries only J_1 .

5.2 The branch point, removed by substitution

The integrand carries $1/k_z$, which is singular at $q = K$. The singularity is integrable but a quadrature driven across it converges only algebraically. Splitting the path at the light circle and substituting

$$q = K \sin \theta \quad (0 \leq \theta \leq \frac{\pi}{2}), \quad q = K \cosh t \quad (t \geq 0), \quad (24)$$

gives, respectively,

$$\frac{q dq}{k_z} = K \sin \theta d\theta, \quad \frac{q dq}{k_z} = -i K \cosh t dt, \quad (25)$$

so the singular factor cancels identically and both pieces are analytic. This is the standard Sommerfeld path. The upper limit $t_{\max}$ is the one convergence parameter; the evanescent branch carries $e^{-K \sinh t |z|}$, so convergence is geometric and $q_{\max}$ of a few times K suffices unless the plane is very close to the sphere.

6 Reflection and transmission

6.1 One-sidedness

Above the sphere there are no sources, so every scattered wave crossing $z = -z_p$ travels in $-z$: the field there is purely up-going. Below $z = +z_p$ it is purely down-going. The representation (17) with $k_z \rightarrow -k_z$ above builds this in, so the spectra of §4 are one-sided by construction and no separation step is required.

6.2 The matrices

Let the incident wave be of type $\iota \in \{P, SV, SH\}$ with unit amplitude, and let $A_\nu^{(\iota)}(\mathbf{k}_\perp)$ be the amplitude of the outgoing wave of type ν obtained from (20) with the Mie coefficients for that incidence, §2.2. Then

$$\mathbf{R}_\nu(\mathbf{k}_\perp) = A_\nu^{(\iota)}(\mathbf{k}_\perp) \Big|_{z=-z_p}, \quad \mathbf{T}_\nu(\mathbf{k}_\perp) = \delta_{\nu\iota} (2\pi)^2 \delta^2(\mathbf{k}_\perp - \mathbf{k}_\perp^0) + A_\nu^{(\iota)}(\mathbf{k}_\perp) \Big|_{z=+z_p}. \quad (26)$$

The delta in $\mathbf{T}$ is the unscattered incident wave (9): it travels in $+z$, so it appears on the lower plane and not the upper one, which is why $\mathbf{R}$ is purely scattered. In practice the delta is kept symbolically and only the scattered part is computed; if a finite lateral grid is used it becomes $\delta_{\mathbf{k}_\perp \mathbf{k}_\perp^0} / \Delta^2$ with Δ the grid spacing in $\mathbf{k}_\perp$.

Both matrices are 3×3 in polarisation at each $\mathbf{k}_\perp$, and both carry the $e^{ik_z z_p}$ phase of the plane they are referred to; changing z_p multiplies them by a diagonal phase and nothing else, which is a useful check.

6.3 State form

If the response is wanted on the six-component state $(\mathbf{u}, \boldsymbol{\tau}_3)$ rather than on mode amplitudes — as it is when comparing against a depth-marching scheme — each mode contributes a column

$$\begin{pmatrix} \mathbf{u} \\ \boldsymbol{\tau}_3 \end{pmatrix}_\nu = \begin{pmatrix} \mathbf{e}_\nu \\ i[\lambda(\mathbf{k} \cdot \mathbf{e}_\nu) \hat{\mathbf{z}} + \mu(k_z \mathbf{e}_\nu + (\mathbf{e}_\nu \cdot \hat{\mathbf{z}}) \mathbf{k})] \end{pmatrix}, \quad (27)$$

obtained by substituting a single plane wave into (13). The six columns — three up, three down — are independent at every $\mathbf{k}_\perp$, so a state on a plane determines its up- and down-going content uniquely. With displacement alone they would not be: six three-vectors in three dimensions are

necessarily dependent, and the traction is what separates the two senses.

7 Summary of the construction

1. Solve the Mie boundary conditions at $r = a$ for a_n , b_n , c_n .
2. Fix $\mathbf{k}_\perp$; take $k_{z,P}$ and $k_{z,S}$ on the branch (2), reversing their sign above the sphere.
3. Evaluate P_n , P'_n and P''_n at k_z/k_P and k_z/k_S — complex arguments, polynomial form — and assemble the angular function and its two surface derivatives, (21).
4. Form the polarisation-resolved amplitudes (20), weight by a_n , b_n , c_n , and sum over n . Both shear families contribute to both transverse polarisations unless $m = 0$.
5. Assemble $\mathbf{R}$ and $\mathbf{T}$ from (26), or the state columns from (27).

No spatial grid, no transform, and no lateral truncation enters at any stage. The only quadrature is the optional reconstruction of §5, whose single parameter is the upper limit of a geometrically convergent integral.

On the azimuthal reduction. The closed forms (23) are stated for $m = 0$, where the only azimuthal dependence is the plane-wave phase. At $m = 1$ the angular function contributes its own $\cos \psi_k$, which multiplies the polarisations' $\cos \psi_k$ and $\sin \psi_k$ and produces J_0 , J_1 and J_2 terms. That bookkeeping is avoidable: the ψ_k integrand is smooth and 2π -periodic, so a uniform rule converges spectrally and the same quadrature serves every m . Nothing in (20) depends on which route is taken — the reduction is only for reconstructing the field, not for forming the response.